



## 1402 – A Comparison of Printing Ink Rub Resistance Using Synthetic and Polyethylene Waxes

### Introduction

Micro Powders offers several grades of synthetic wax and polyethylene wax to the coatings and inks markets. These finely micronized additives will improve rub resistance, with the level of improvement a function of the nature of the polymer used. The following wax additives will be compared:

| Product:  | Description:           | Mean particle size, $\mu\text{m}$ : | Maximum particle size, $\mu\text{m}$ : |
|-----------|------------------------|-------------------------------------|----------------------------------------|
| MPP-230VF | Lower MW polyethylene  | 11 maximum                          | 31                                     |
| MPP-620F  | Medium MW polyethylene | 7-9                                 | 31                                     |
| MPP-635F  | Higher MW polyethylene | 8-10                                | 31                                     |
| MP-22     | Synthetic Wax          | 8-10                                | 31                                     |

### Objective

Compare the efficiency in rub resistance improvement with a range of wax additives having similar particle size distribution but different wax chemistry.

### Study Methodology

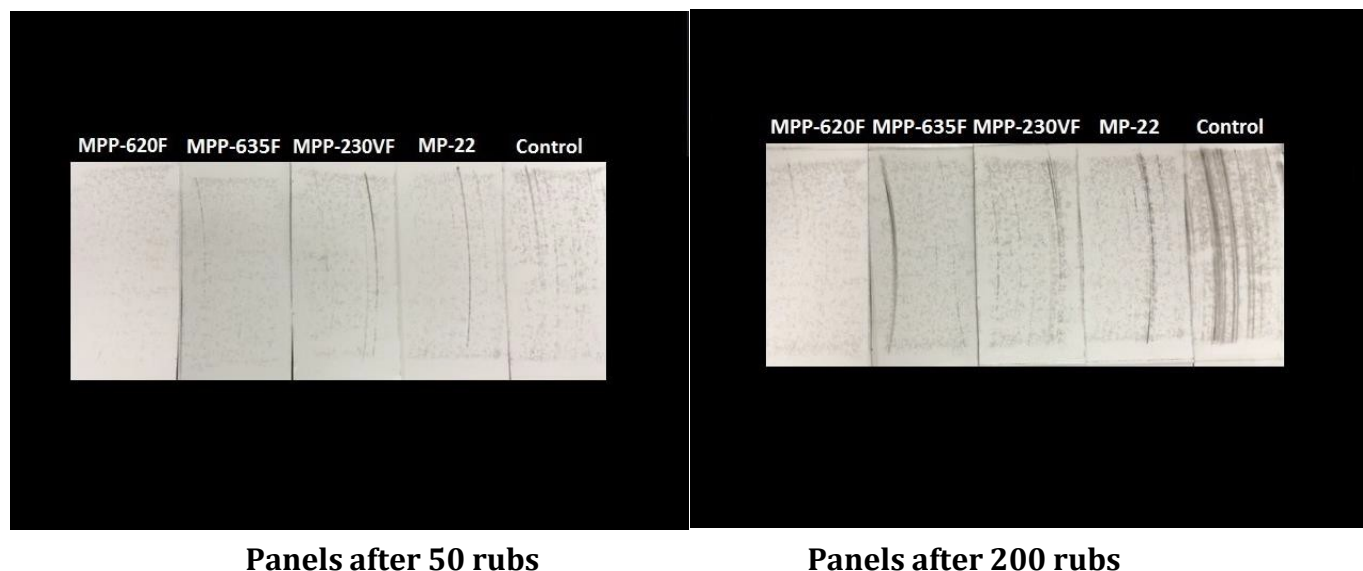
The test formula for this study was an acrylic waterbased black printing ink that contained no wax additives. Each wax additive was dosed and fully dispersed at a 1% level based on total formula weight. The modified ink samples were drawn down on sealed Leneta cards using a #10 wire wound rod and allowed to dry for 1 week at ambient conditions before testing. A #10 wire wound rod applies a wet film thickness of approximately 0.01" (25.6  $\mu\text{m}$ ).

A Sutherland Rub tester was used to abrade the test panels. The tester is a motor driven instrument that moves a weighted test strip over a printed specimen through an arc. The 4 pound (1.8 kg) weight was used. Each panel was tested on the Sutherland by abrading the printed surface against white unsealed Leneta card. Panels were abraded for 50 and 200 cycles.



## Results

The panels were rated on a 1 to 5 scale, where a rating of 1 indicates the least amount of visible ink transfer and a rating of 5 indicates the most amount of visible ink transfer.



The results were ranked as follows (5 = worst, 1 = best):

|          | MPP-620F | MPP-635F | MPP-230VF | MP-22 | Control |
|----------|----------|----------|-----------|-------|---------|
| 50 rubs  | 1        | 2        | 3         | 4     | 5       |
| 200 rubs | 1        | 2        | 3         | 4     | 5       |

## Conclusions

- MPP-620F was the best overall performer, with the least amount of ink transfer (and therefore the highest improvement in rub resistance). This product is based on a unique grade of polyethylene and is designed to provide maximum rub resistance in this class of additive.
- MPP-635F, based on a higher molecular weight polyethylene, was rated second for rub resistance.
- MPP-230VF, an economical grade based on a lower molecular weight polyethylene, was rated third.
- MP-22, based on synthetic wax, was the worst performer in the study (although it did provide some improvement in rub resistance vs. the control). This additive is intended more for slip and lubricity than for rub and mar resistance so these results are not unexpected.
- All Micro Powders' additives tested did improve rub resistance when compared to the unmodified control.

Although this study was conducted in a waterbased printing ink formula, the results and differences in performance are translatable to their use in other wax-modified surface coatings.

### Micro Powders, Inc.

580 White Plains Road,  
Tarrytown, NY 10591  
TEL 914.793.4058 FAX 914.472.7098  
Email: info@micropowders.com

\*The above data reflects typical properties. Please contact Micro Powders for official product specifications. The information contained herein is to the best of our knowledge true and correct and any suggestions are made without guarantee, express or implied, since conditions of use are beyond our control. Micro Powders, Inc. disclaims any liability incurred in connection with the use of any data or suggestions. Nothing contained herein shall be construed as a recommendation to infringe on any existing patents covering any material or its use.

Certified to ISO 9001